

2022 Ph H1 Q8

Section: Our Dynamic Universe

Topic: The Expanding Universe

A student makes the following statements about the expanding Universe:

- I. The evidence supporting the existence of dark matter comes from estimations of the mass of galaxies.
- II. The evidence supporting the existence of dark energy comes from the accelerating rate of expansion of the Universe.
- III. The peak wavelength of radiation emitted by hotter stars is longer than that for cooler stars.

Which of these statements is/are correct?

Evaluation:

-  I is correct: orbital speeds of stars suggest more mass than observed → **dark matter**
 -  II is correct: observations of supernovae and cosmic expansion support **dark energy**
 -  III is false: **hotter** stars have a **shorter** peak wavelength (Wien's Law)
-

Final Answer:

D

Revision Tips:

- **Dark matter** is inferred from galaxy rotation curves and gravitational lensing
- **Dark energy** explains why the expansion of the Universe is accelerating
- **Wien's Law:** hotter → shorter peak wavelength

$$\lambda_{\max} \propto \frac{1}{T}$$